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J. Chem. Phys. 143, 014501 (2015)

<https://doi.org/10.1063/1.4926394>



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A modified two-state empirical valence bond model for proton transport in aqueous solutions

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(Received 11 March 2015; accepted 25 June 2015; published online 6 July 2015)

A detailed analysis of the proton solvation structure and transport properties in aqueous solutions is performed using classical molecular dynamics simulations. A refined two-state empirical valence bond (*a*TS-EVB) method, which is based on the EVB model of Walbran and Kornyshev and the anharmonic water force field, is developed in order to describe efficiently excess proton transport via the Grotthuss mechanism. The new *a*TS-EVB model clearly satisfies the requirement for simpler and faster calculation, because of the simplicity of the two-state EVB algorithm, while providing a better description of diffusive dynamics of the excess proton and water in comparison with the previous two-state EVB models, which significantly improves agreement with the available experimental data. The results of activation energies for the excess proton and water calculated between 300 and 340 K (the temperature range used in this study) are also found to be in good agreement with the corresponding experimental data. © 2015 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4926394>]

I. INTRODUCTION

Because of the importance of water as a solvent in chemical, biochemical, and biophysical processes, many experimental and theoretical research efforts have focused on understanding its properties in terms of the nanoscopic structure and dynamics.¹ Proton transport (PT) through an aqueous phase, such as in acid solutions, hydrated polymer electrolyte membranes, or proteins, is also a significant dynamic process in many different fields of chemistry and biology.

In a bulk aqueous solution at infinite dilution, proton mobility is roughly five times higher than that of other cations, such as K⁺, which have an ionic radius similar to hydronium ions.² This property of PT is attributed to a combination of the proton hopping mechanism, known as the Grotthuss mechanism,² and the vehicular mechanism. In the Grotthuss mechanism, a proton diffuses through the hydrogen bond network of water molecules, whereas in the latter mechanism, a proton diffuses in solution as a hydronium ion, which is the smallest stable hydrated form of an excess proton in water. While PT via the Grotthuss mechanism is believed to involve the interconversion between two different solvation structures, the Eigen³ ([H₃O·(H₂O)₃]⁺) cation and Zundel⁴ ([H·(H₂O)₂]⁺) cation,^{5–7} details of the hopping mechanisms are still under debate.

Theoretically, a number of water force fields have been developed for molecular dynamics (MD) simulations, including atomistic models based on rigid bonds^{8–15} and flexible bonds.^{16–19} With these models, the structural, thermodynamic, and dynamic properties of water, including hydrogen bond structure and dynamics,^{20–25} have been successfully reproduced to a certain degree. However, because of the nu-

clear quantum effects caused by the small mass of protons and the delocalization of the excess charge over multiple water molecules, the accurate theoretical modeling of PT in aqueous solution remains a challenge. The precise physics of PT can in principle be captured in *ab initio* simulations.^{5–7,26–30} However, extensive simulations using these methods for a larger number of atoms, where systems reproduce bulk behavior with reliable statistics, have been limited because of the high computational cost. A central force model^{31,32} for PT has also been developed explicitly to take into account dissociation of a proton from the hydronium ion in the MD simulations, which provides an insight into the structural and dynamic properties of the excess proton with less demanding computations than *ab initio* simulations.

As the most effective reactive force fields for the molecular interactions, empirical valence bond (EVB) approaches, such as the two-state EVB (TS-EVB) method,^{33,34} the multistate EVB (MS-EVB) method,^{35–39} and the self-consistent multistate EVB (SCI-MS-EVB) method,⁴⁰ have been extensively developed to allow PT phenomena to be simulated accurately in MD simulations. These methods have been built upon the development of the EVB method originally formulated by Warshel and Weiss⁴¹ and subsequently investigated by Lobaugh and Voth.⁴² Although the MS-EVB method can be used to study multiple protons, the requirement of as many as 40 EVB states per proton and the exponential growth of their matrix size with respect to the number of excess protons practically prohibit the model's use in multiproton systems. In turn, the SCI-MS-EVB method⁴⁰ has been developed to capture the essential features of PT in multiproton systems, providing in-depth insight into proton solvation and transport properties in aqueous systems. However, because of the requirement of a large number of EVB states, as with the MS-EVB method, the efficient application of the model in large scale multiproton environments is still limited to a certain degree, although the

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computational cost of the method scales linearly with respect to the number of protons, and various multiproton systems have been studied using the model.^{43–49}

Alternatively, the TS-EVB method has also been developed, providing an essential simplification of the PT algorithm in comparison with the multistate EVB framework, which clearly satisfies the requirement of faster calculation and is easily applicable to longer simulations and larger systems in multiproton environments, such as polymer electrolyte membranes,^{50–52} where large atomistic or coarse-grained MD simulations are required to observe the extensive morphology of the systems. Furthermore, the TS-EVB model can also be applied effectively to confined water structure environments such as water wires in narrow pores, where the interconversion of proton transferring clusters is limited to a Zundel complex, which is different in many ways from PT in bulk water. The Walbran and Kornyshev³⁴ (WK) version of the TS-EVB method describes proton transport through a sequence of interconversion between Zundel and Eigen complexes and allows the efficient and energy-conserving simulation of multiproton systems. The model does, however, have some caveats. In particular, only one water molecule—the closest to the hydronium ion—can be assigned as a partner where the proton can be transferred during the transferring process at any given time, while there can be two additional neighboring water molecules, forming an Eigen cation, which may become more energetically favorable candidates for proton transfer during the transferring process with the first partner water molecule. Furthermore, this shortcoming restricts a charge delocalization effect beyond the partner cluster of the Zundel cation, which may be particularly important for calculating vibrational spectra.³⁸ In the WK model, a high-frequency and inertialess electronic polarizable water model is employed, and the dynamical changes of the EVB energy profile (i.e., the dynamical changes in the electronic interaction) due to the inertialess dynamics of the point charges result in reducing the Franck–Condon barriers and yielding an activation barrier compatible with the experimental one, although the inertialess polarizability should not directly contribute to lowering the activation barrier in terms of physics because electronic degrees of freedom are inertialess. The authors have targeted their model in their efforts to understand the effect of confined heterogeneous environments on the PT properties in their later studies.^{33,53} However, the WK model, with its underlying polarizable water model, predicted a pre-exponential factor of proton mobility half of that which is obtained experimentally and an excessive degree of water diffusion. The latter is caused by the discrepancy in the water–water structure with the experimental data, resulting in the diffusion coefficient of water being almost three times higher than that of real water.

In this study, in order to provide a more quantitative description of PT mobility, a new model, which is a modified version of the WK model, was developed. The nonpolarizable anharmonic water force field (*a*SPC/Fw),³⁷ in which the vibrational anharmonicity of the underlying water model is introduced into the original nonpolarizable harmonic SPC/Fw water model,¹⁹ was employed in the new model as an alternative to the original polarizable water model, providing a reasonable representation of the OH intramolecular vibrations

in a case where nuclear quantum effects are explicitly taken into account.⁵⁴ The introduction of the anharmonic effects into the original WK model leads to a new anharmonic TS-EVB (*a*TS-EVB) model. Furthermore, as the PT in aqueous solutions involves fast proton transfer between two water molecules on the subpicosecond scale, an accurate description of the potential energy surface (PES) is essential for more reliable description of the PT properties. Thus, unlike the WK model, the new *a*TS-EVB model was parameterized to reproduce the PESs calculated by the *ab initio* calculations for a reliable description of the energy barrier at each molecular geometry. In this regard, our model implicitly suppresses to some extent the contribution from the Franck–Condon effects to the activation energy without explicit inclusion of polarizability. Consequently, the *a*TS-EVB model has been refined fully in terms of the proton–water and water–water structures, the absolute values of proton and water mobility, the proton/water diffusion coefficient ratios, and the activation energy for protons and water. The *a*TS-EVB model efficiently provides a more accurate description of the structural and dynamic properties of both the solvated protons and water in concentrated acid solutions and significantly improves agreement with available experimental data.

This paper is organized as follows. Section II provides details of the simulation methods, including the *a*TS-EVB calculation algorithm, the parameterization of the *a*TS-EVB model, and the computational details. Section III discusses the simulation results for the analysis of proton solvation and transport properties. Section IV summarizes the findings and presents our conclusions.

II. SIMULATION METHODS

A. *a*TS-EVB model

The two-state EVB approach, which is based on the WK model,³⁴ was used in this study to capture efficiently the underlying physics of multiproton systems within the limits of available computational capabilities. In the *a*TS-EVB model, we considered a simplified cluster model of the Zundel cation where a hydronium ion exchanges a proton with the closest neighboring water molecule. Within a single cluster, we considered a function of the proton transfer coordinate, defined as follows:

$$Q \equiv |R_{O^*H}| - |R_{OH}|, \quad (1)$$

where O^* and O are the oxygen atoms of a hydronium ion and a water molecule, respectively. Following Ref. 34, the charge switching between the hydronium ion and its partner water molecule takes place on each of the oxygen sites (O^* and O) and the hydrogen sites (H^* and H), according to the following expressions:

$$\tilde{q}_{O^*} = q_{O^*} [1 - f(Q)] + q_O f(Q), \quad (2)$$

$$\tilde{q}_O = q_O [1 - f(Q)] + q_{O^*} f(Q), \quad (3)$$

$$\tilde{q}_{H^*} = q_{H^*} [1 - f(Q)] + q_H f(Q), \quad (4)$$

$$\tilde{q}_H = q_H [1 - f(Q)] + q_{H^*} f(Q), \quad (5)$$

where q_{O^*} , q_O , q_{H^*} , and q_H are the point charges of corresponding atoms in the classical forms, and $f(Q)$ is the charge

switching polynomial as a function of Q , defined in Ref. 34 as

$$f(Q) = \frac{1}{2} + \frac{\alpha}{5} \left(\frac{Q}{Q_0} \right)^5 - \frac{2\alpha}{3} \left(\frac{Q}{Q_0} \right)^3 + \alpha \left(\frac{Q}{Q_0} \right), \quad (6)$$

with $\alpha = 15/16$, to obtain the ± 1 limits at $Q/Q_0 = \pm 1$. Within the cluster, a reactive PES, V , can be modeled by an EVB approach, in which the reactant and product valence bond configurations, Ψ_1 and Ψ_2 , interact via an empirical Hamiltonian, $\mathbf{H}$,

$$\Psi = c_1 \psi_1 + c_2 \psi_2, \quad (7)$$

$$\mathbf{H} = \begin{bmatrix} V_i & \Lambda \\ \Lambda & V_f \end{bmatrix}, \quad (8)$$

$$V_i = \langle \psi_1 | \hat{H} | \psi_1 \rangle, \quad (9)$$

$$V_f = \langle \psi_2 | \hat{H} | \psi_2 \rangle, \quad (10)$$

$$\Lambda = \langle \psi_1 | \hat{H} | \psi_2 \rangle, \quad (11)$$

$$V = \frac{(V_i + V_f) - \sqrt{(V_i - V_f)^2 + 4\Lambda(Q, R)^2}}{2}. \quad (12)$$

Here, i and f stand for the “initial” and “final” states of the cluster during the transferring process, respectively. In the initial state, V_i is the energy of the cluster, calculated under the assumption that the molecule “ H_3O^+ ” (left molecule) is treated as the hydronium model, while the molecule “ H_2O ” (right molecule) is considered as the water model, as schematically shown in Figure 1(a). In the final state, with the same configuration as that of the initial state, the left molecule is now separated from the central proton and treated as the water model, whereas the right molecule is combined with its proton and considered as the hydronium model, as shown in Figure 1(b). Note that both V_i and V_f are calculated at the same position of the central proton at each step during the transferring process. The term $\Lambda(Q, R)$ is an interaction matrix element and is assumed to have the following form:

$$\Lambda(Q, R) = \left\{ \left(\frac{Q}{Q_0} \right)^4 - 2 \left(\frac{Q}{Q_0} \right)^2 + 1 \right\} F(Q, R), \quad (13)$$

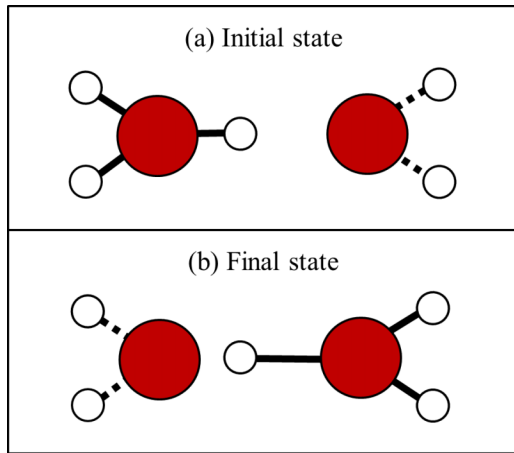


FIG. 1. (a) Initial state and (b) final state of the H_5O_2^+ complex at the same position of the central proton. For each state, interactions of the hydronium ion are indicated by solid lines and those of the water molecule are indicated by dashed lines.

where the term $F(Q, R)$ is a polynomial function, given by

$$F(Q, R) = \alpha(R) \cdot \left(\frac{Q}{Q_0} \right)^4 + \beta(R) \cdot \left(\frac{Q}{Q_0} \right)^2 + \gamma(R), \quad (14)$$

$$\alpha(R) = \alpha_1 \cdot R^3 + \alpha_2 \cdot R^2 + \alpha_3 \cdot R + \alpha_4, \quad (15)$$

$$\beta(R) = \beta_1 \cdot R^3 + \beta_2 \cdot R^2 + \beta_3 \cdot R + \beta_4, \quad (16)$$

$$\gamma(R) = \gamma_1 \cdot R^3 + \gamma_2 \cdot R^2 + \gamma_3 \cdot R + \gamma_4. \quad (17)$$

Here, Q is the proton transport variable defined in Eq. (1), and R is the distance between O^* and O . Q_0 is a constant parameter that determines the width of the matrix element. The function $\Lambda(Q, R)$ smoothly goes to value 0 for $Q \leq -Q_0$ and $Q \geq Q_0$, where the total potential energy of the Zundel-like structure V as shown in Eq. (12) is equal to V_i and V_f , respectively. Generally, each of the three water molecules, which normally form the solvation shell of the hydronium ion, is a possible receptor of a proton from a hydronium ion. In the model, we assume that the water molecule closest to the hydronium ion is the most likely candidate for a proton transferring partner. However, partnering with a water molecule, when the dynamically changing values of Λ as a function of Q are nonzero, i.e., $Q > -Q_0$, would lead to a discontinuous interaction potential and a loss of energy conservation. Therefore, the partnering process was restricted to the times only when $Q \leq -Q_0$ to retain a continuous total potential energy of the cluster. The constant length scale Q_0 was adjusted to be short enough not to limit the occurrence of the partner switching (i.e., the interconversion between Zundel and Eigen complexes) at $Q \leq -Q_0$. Consequently, a 20 times higher frequency of the partner switching than that of the hopping rate, i.e., an average of 20 times partner switching per proton hopping, was observed in our model. It should be noted that the present model implicitly takes into account the $\text{O}^*-\text{H}-\text{O}$ angle effect on the cluster configuration to be energetically favorable, although the variable Q as in Eq. (1) is only defined by the distance information, because the total EVB energy V in Eq. (12) will become quite high because of the high cluster energy for both V_i and V_f if the geometry of the cluster is not a favorable configuration.

In the case of multiproton environments, the “local” nature of the charge switching function algorithm in our model, as with the WK model, allows the EVB complexes for each proton to be handled independently, unless two hydronium ions share the same neighboring water molecule as a transferring partner. Practically, this assumption can be rationalized when the proton concentration is sufficiently low, so that a certain distance between hydronium ions can be retained at any spatial distribution, where a water molecule is unlikely to be the closest one for two hydronium ions. Therefore, a linear scaling of the computational cost with respect to the number of protons can be achieved using the $a\text{TS-EVB}$ model for the efficient simulation of multiproton systems.

B. Parameterization of the $a\text{TS-EVB}$ model

The $a\text{TS-EVB}$ was parameterized using the *ab initio* calculation to provide an accurate description of both the dynamic and solvation properties of the excess proton. The PES of the proton shuttling between two water molecules

TABLE I. Parameters for the EVB model.

Q_0	0.5						
α_1	-9591.4	α_2	72 619.4	α_3	-182 837.0	α_4	152 923.5
β_1	708.2	β_2	-5 265.8	β_3	13 122.5	β_4	-10 930.0
γ_1	-83.2	γ_2	677.6	γ_3	-1 790.8	γ_4	1 585.7

in the Zundel configuration was calculated by scanning the central hydrogen between two water oxygen atoms at O*–O separations varying from 2.3 to 2.6 Å, while all other atoms were allowed to relax. Although in the MS-EVB models the PES is generally calculated for O*–O separations varying from 2.2 Å to 2.8 Å in increments of 0.2 Å,^{35–37} in our model, a more detailed calculation at the distance between 2.3 Å and 2.6 Å in increments of 0.1 Å was performed to reproduce the PES more accurately in the energetically favorable range of $-Q_0 < Q < Q_0$. The geometries of the Zundel configuration were optimized using density functional theory with the 6-31G** basis set, and B3LYP⁵⁵ was employed for the exchange correlation functional. The geometry optimizations were followed by the total energy calculation of the optimized geometries using the single, double, and perturbative triple excitation coupled-cluster method with the cc-pVTZ basis set to obtain more accurate proton shuttling PESs. In all situations, spin multiplicity was set to singlet. All the *ab initio* calculations were carried out with Gaussian 09.⁵⁶ The results were found to be similar to those calculated by Park *et al.*,³⁷ where a similar calculation method was used except for the use of the MP2/aug-cc-pVTZ level of theory for the geometry optimization of the Zundel cation. The final set of parameters for the *a*TS-EVB model is given in Table I.

Figure 2 shows a comparison between the *ab initio* data and the results obtained with the *a*TS-EVB model for O*–O separations corresponding to 2.3 Å, 2.4 Å, 2.5 Å, and 2.6 Å. In all cases, the results with the *a*TS-EVB model are in good agreement with the corresponding *ab initio* data. Note that although the largest energy difference between the *ab initio* and the EVB curves, ~4.0 kcal/mol, was found at $Q = \pm 0.5$

for O*–O separations of 2.3 Å, this discrepancy exerts little influence on the proton shuttling pathway. For O*–O separations of 2.4 Å, 2.5 Å, and 2.6 Å, the overall curves of the *a*TS-EVB model are in good agreement with those of the *ab initio* calculations, where the largest energy difference between them is ~0.7 kcal/mol at $Q = \pm 0.4$ for O*–O separations of 2.5 Å.

C. Computational details

The EVB simulations of protonated water were carried out using classical MD simulations. The system consisted of the nonpolarizable *a*SPC/Fw model³⁷ of 215 water molecules, one hydronium ion, and one chloride (Cl[−]) counterion in a periodic cubic box, which corresponds to the proton concentration of a 0.26M HCl solution. The atomic partial charges of the water molecule and hydronium ion (−0.835*e* for water oxygen, +0.4175*e* for water hydrogen, −0.5*e* for hydronium oxygen, and +0.5*e* for hydronium hydrogen) were taken from the *a*SPC/Fw model.³⁷ However, in this model, the charges of the hydronium ion were scaled by 0.7, i.e., total charges of +0.7*e* (−0.35*e* for hydronium oxygen and +0.35*e* for hydronium hydrogen) and of −0.7*e* for a chloride anion were used. The damped charge model of the hydronium ions was employed to bring the formation energies of the Zundel cation into line with the *ab initio* calculations. As the nonpolarizable water models (e.g., the *a*SPC/Fw model) that are parameterized to reproduce the bulk phase dipole moment overestimate the electrostatic interaction between a hydronium ion and a water molecule in the Zundel cation, it is reasonable to use the damped point charges for the hydronium ions. A similar approach was also used in the previous MS-EVB models^{38,57} to compensate for the overestimation of the Coulomb interaction between the solvated ions and water molecules. Furthermore, an investigation of Zhao *et al.*⁵⁸ that used MD simulations, using the polarizable force fields, showed that a charge transfer of 0.2–0.3*e* from the chloride ion to the neighboring water molecules takes place in the system of a single chloride ion in bulk water. Similarly, a significant charge transfer was observed in *ab initio* simulation studies of an ionic liquid, where a charge between −0.7*e* and −0.8*e* was reported for the chloride anion.⁵⁹ However, as the current model restricts transfer of the full formal ionic charge, an accurate estimation of the polarization response to a solvated ion without using the damped charge model by applying the polarizable multipole water model,^{60–62} which may also reduce the overestimation of the Coulomb interactions, is desirable to reproduce the full-charge transition process between the two. Such a significant modification of incorporating electronic polarization effects into the present model, in a way that is consistent with the proton shuttle mechanism, will therefore be a focus for future *a*TS-EVB model development.

The parameters of the Lennard-Jones (LJ) 12–6 interactions for the chloride ($\epsilon = 0.5216$ kJ/mol and $\sigma = 3.785$ Å) were taken from Koneshan *et al.*⁶³ and those for the hydronium ion and the water molecule were taken from the *a*SPC/Fw model.³⁷ The Lorentz–Berthelot mixing rules were used for intermolecular interactions. After all molecules were added, systems were constructed at $T = 298.15, 320,$ and 340 K.

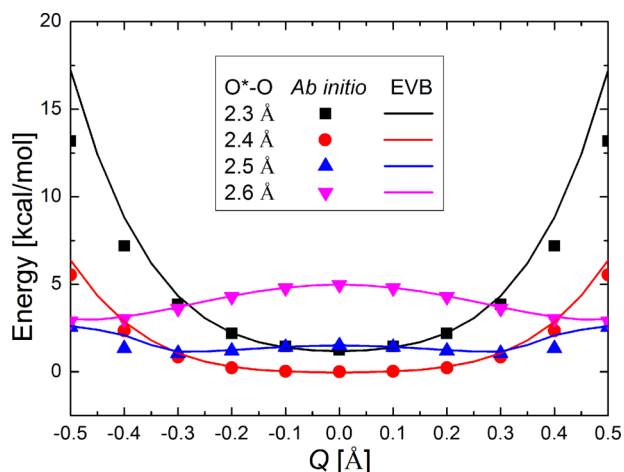


FIG. 2. PES curves for the proton shuttling in the Zundel cation at O*–O distances of 2.3, 2.4, 2.5, and 2.6 Å. Symbols denote results calculated from the *ab initio* data and lines denote results calculated from the EVB model.

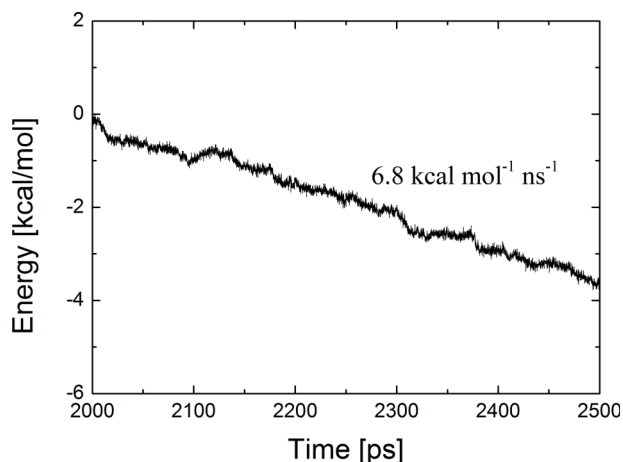


FIG. 3. Total system energy as a function of the time for *a*TS-EVB model in constant NVE simulations.

All the systems were first equilibrated for 1 ns at a constant pressure of 1 atm and temperatures of 298.15, 320, and 340 K, respectively. Subsequently, the systems were further equilibrated in the canonical NVT ensemble for 1 ns. The structural and dynamic properties were then calculated from a 1 ns long simulation without the constant temperature algorithm, i.e., in the constant NVE ensemble. For each system, the trajectory of molecules was collected every 0.1 ps. The equations of motion were integrated using the Verlet algorithm with a time step of 1 fs. The temperature was held constant by a Nosé–Hoover thermostat^{64,65} and the pressure was controlled by the Andersen method.⁶⁶ The LJ interactions were treated using the spherical cutoff method, with a cutoff of 9 Å. The smooth particle mesh Ewald method⁶⁷ with a precision of 10^{-6} was used to calculate electrostatic interactions.

Figure 3 shows the total system energy for the *a*TS-EVB model in the NVE simulations at $T = 298.15$ K. A constant energy drift of $6.8 \text{ kcal mol}^{-1} \text{ ns}^{-1}$ was found, which is between $3.0 \text{ kcal mol}^{-1} \text{ ns}^{-1}$ and $13.1 \text{ kcal mol}^{-1} \text{ ns}^{-1}$ for the *a*MS-EVB3 and MS-EVB2 models, respectively. However, the effect of the energy drift on the proton transfer should not be significant because the energy fluctuation as a function of time in the figure is relatively small in comparison with the PES barrier ($\sim 0.5 \text{ kcal mol}^{-1}$) observed in the free energy profile. The discrepancy in energy conservation is mainly due to the potential energy function used in the models, which should change smoothly with no loss of the total energy during a proton transfer. The precise energy conservation properties of the *a*TS-EVB model still remain to be determined. Specifically, the improvement of the function $\Lambda(Q, R)$ is a key factor leading to better energy conservation, while the other properties are maintained, which is a challenge for the future, to refine further the *a*TS-EVB model.

III. RESULTS AND DISCUSSION

A. Structural properties

The water and proton solvation structure can be characterized by the radial distribution functions (RDFs). First, the

RDFs between water molecules, calculated in the simulation at $T = 298.15$ K, were compared with the available experimental data extracted from neutron and X-ray scattering measurements of the water structure factor⁶⁸ and the corresponding RDFs for the WK model, as shown in Figure 4. In all cases, the simulated results for the *a*TS-EVB model are in good agreement with the experimental results as well as with the original *a*SPC/Fw water structures in bulk water³⁷ and comparable to

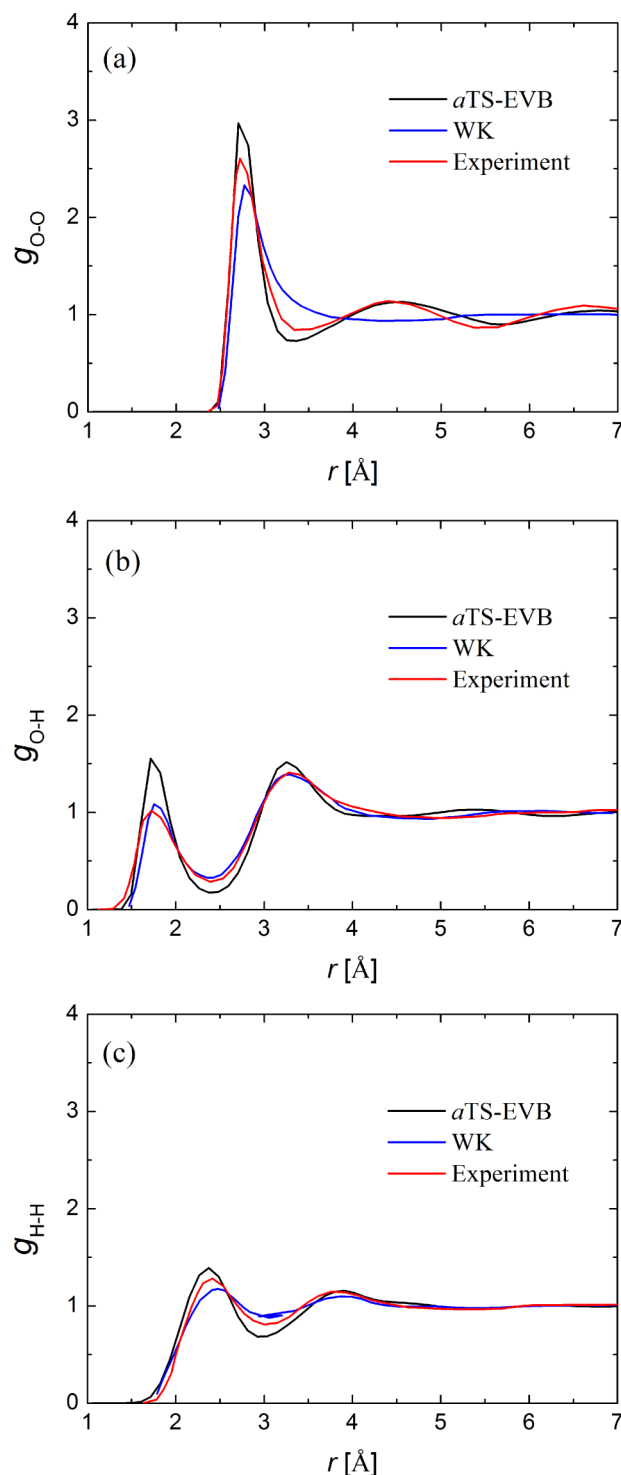


FIG. 4. RDFs of water molecules between (a) oxygen atoms (O–O), (b) oxygen atom and hydrogen atom (O–H), and (c) hydrogen atoms (H–H) at 298.15 K. The experimental data are from Ref. 68.

the WK model, except that the WK model shows a broadened first peak for the water oxygen–oxygen structure, which is a sign of overestimation for the WK model’s water mobility.

The RDFs between the hydronium atoms (O_h and H_h) and the water atoms (O and H) were calculated at $T = 298.15$ K. Figure 5 shows their comparison with the corresponding experimental data.⁶⁹ In the EVB calculation, the oxygen atom closest to the transferring proton is defined as O_h and the hydrogen atoms belonging to the O_h atom, including the transferring proton, are considered to be H_h , i.e., the O^* atom is defined as O_h if $Q < 0$ and a partner water oxygen atom (O) is defined as O_h if $Q \geq 0$. The RDF of the water oxygen atoms around the hydronium oxygen atom (O_h –O) obtained with the WK model is also shown in Figure 5(a) for comparison. The results for the a TS-EVB model are similar to the corresponding RDFs obtained with the MS-EVB models,^{35,37} except that g_{O_h-O} and g_{H_h-O} for the a TS-EVB model both show a lower first peak in comparison with the MS-EVB models. This may be because of the simplicity of the EVB formulation limiting the simultaneous fractional charge and potential mixing with multiple water molecules in the Eigen cation. Nevertheless, the agreement with experimental data is significantly improved in comparison with the results obtained in the WK model. Specifically, the first valley of g_{O_h-O} for the a TS-EVB model located at ~ 2.9 Å is in better agreement with the experimental value of ~ 2.8 Å, which correctly shows the first solvation shell of three water molecules around the hydronium ion (i.e., the Eigen cation), while that for the WK model shows a broader valley located at ~ 3.8 Å, as shown in Figure 5(a). This result suggests that the WK model has weaker interaction of a hydronium ion with water molecules, which may also cause the significantly higher water mobility than that of real water, with the result that there are large discrepancies for the water diffusion coefficient and the proton/water diffusion coefficient ratios in comparison with the experimental data. From these aspects, significant improvement over the WK model has been achieved; the proton–water and water–water structures of our model are in better agreement with the experimental data. A comparison of the detailed dynamic properties in our model with those of the WK model is discussed in Section III B.

B. Dynamic properties

To calculate the proton self-diffusion coefficient with its continuous trajectory, the position vector of the proton, whose dynamics involves both the vehicular (or continuous) motion and the Grotthuss (or discontinuous) motion, is defined in Ref. 34 as

$$\mathbf{r}_{H^+} = \mathbf{r}_{O^*} [1 - f(Q)] + \mathbf{r}_{O^*} f(Q), \quad (18)$$

where $\mathbf{r}$ is the position vector of corresponding atoms, Q is the proton transfer coordinate defined in Eq. (1), and $f(Q)$ is the charge switching function defined in Eq. (6). The self-diffusion coefficients of the proton D_{H^+} can be calculated from the mean square displacements (MSDs) using the Einstein–Smoluchowski equation,

$$D = \lim_{t \rightarrow \infty} \frac{1}{6t} \langle |\mathbf{r}_{H^+}(t) - \mathbf{r}_{H^+}(0)|^2 \rangle. \quad (19)$$

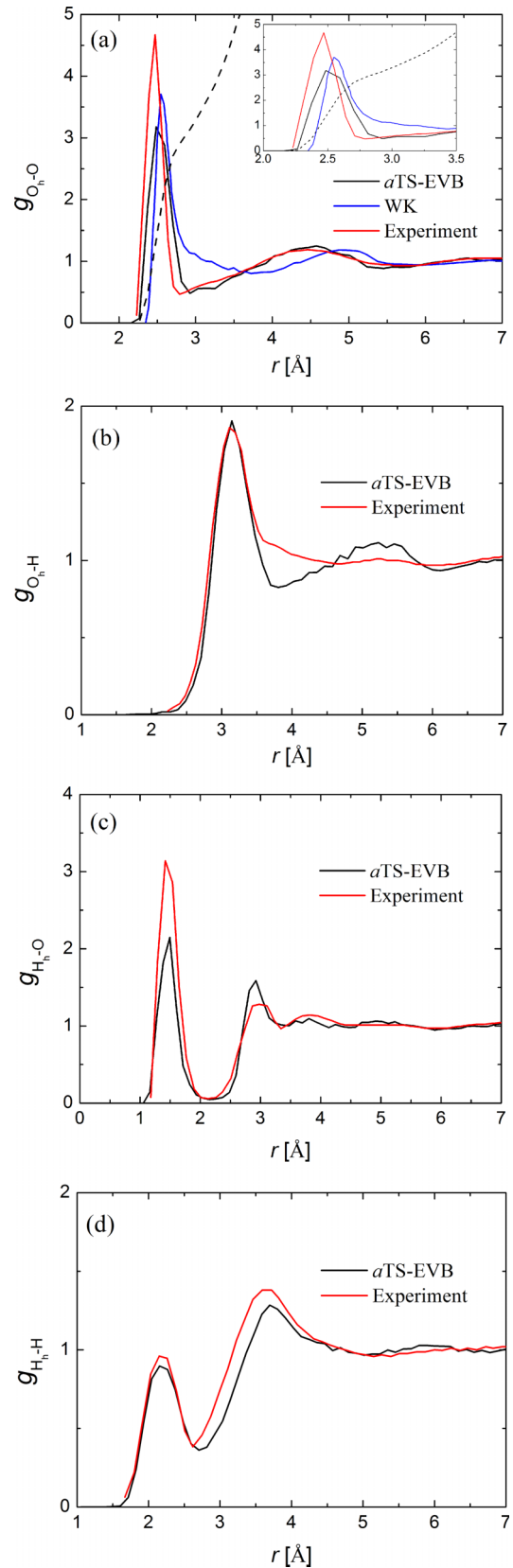


FIG. 5. RDFs of water atoms (O and H) around the hydronium atoms (O_h and H_h) calculated at 298.15 K: (a) RDF and coordination number (solid and dashed lines) for water oxygen atoms around the hydronium oxygen atom, (b) RDF for water hydrogen atoms around the hydronium oxygen atom, (c) RDF for water oxygen atoms around the hydronium hydrogen atom, and (d) RDF for water hydrogen atoms around the hydronium hydrogen atom. The experimental data are from Ref. 69. The inset in (a) is a magnified view of the first peak of g_{O_h-O} .

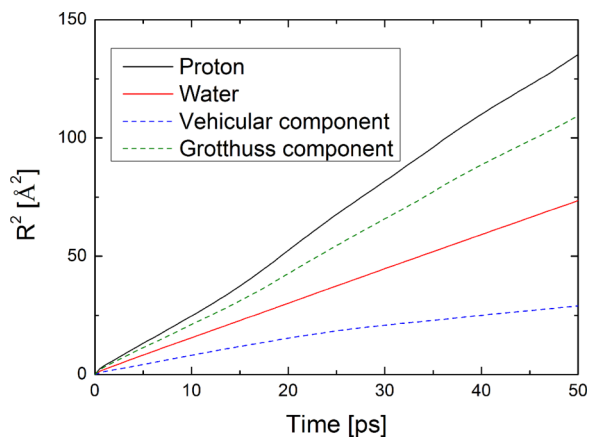


FIG. 6. MSDs of the proton (black), the water molecule (red), and the vehicular (blue) and Grothuss (green) components of the proton MSD at 298.15 K as a function of time.

The MSDs for the proton and water at $T = 298.15$ K are depicted in Figure 6. The proton self-diffusion coefficient, estimated by the corresponding MSD slope from 15 to 50 ps, is $D_{H^+} = 4.9 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$, which is $\sim 20\%$ and $\sim 69\%$ higher than the corresponding values obtained with the *a*MS-EVB3 and MS-EVB3 models, respectively. This may be because reducing the total charge of the hydronium ions reduces the need to consider the explicit polarization effects. However, although significant improvement over the previous models was found, the D_{H^+} calculated with our model is still somewhat lower than the experimental value of $9.4 \times 10^{-5} \text{ cm}^2/\text{s}$ at infinite dilution.⁷⁰ Furthermore, given the concentration of the system used in the simulation, the proton self-diffusion coefficient at the concentration of 0.26M HCl should differ from the value obtained experimentally at infinite dilution because of the presence of the counterions. The experimental proton diffusion coefficient in 0.26M HCl solution can be predicted from the experimentally determined equivalent conductance Λ' of HCl at 0.26M, which is $37.8 \text{ mS m}^2 \text{ mol}^{-1}$.⁷¹ Using the relation $D_{H^+} = t_{H^+} \Lambda' RT / z_{H^+}^2 F^2$, where t_{H^+} is the transport number of 0.82 at 298 K, which is nearly constant,³¹ z_{H^+} is the valence of the proton, and F is the Faraday constant, the experimental value of $8.2 \times 10^{-5} \text{ cm}^2/\text{s}$ for the proton at 0.26M can then be obtained. The value of the self-diffusion coefficient in the *a*TS-EVB model is now found to be within $\sim 40\%$ of the experimental value. Although the WK model³⁴ also shows the similar value of $4.7 \times 10^{-5} \text{ cm}^2/\text{s}$ for the proton diffusion coefficient, this value should be discussed in association with the relation to the underlying water diffusion using the proton/water diffusion coefficient (D_{H^+}/D_{H_2O}) ratios, as detailed below. Note that because of the simplified classical nature of the model, nuclear quantum effects, which may affect the physics of proton diffusion to a certain degree, have not been explicitly considered here.

The water self-diffusion coefficient is $D_{H_2O} = 2.4 \times 10^{-5} \text{ cm}^2/\text{s}$, which is in good agreement with the experimental value⁷² of $2.3 \times 10^{-5} \text{ cm}^2/\text{s}$. The results show a significant improvement of water mobility, particularly in comparison with the value calculated with the WK model³⁴ of $6.3 \times 10^{-5} \text{ cm}^2/\text{s}$, whereas the proton diffusion coefficient obtained from the *a*TS-EVB model indicates a similar value for

the WK model. As the proton diffusion coefficient is strongly affected by the water diffusion coefficient, it is thus more reasonable to compare the D_{H^+}/D_{H_2O} ratios. The D_{H^+}/D_{H_2O} ratios are 0.75, 2.0, and 3.6 for the WK model, the *a*TS-EVB model, and the experimental value, respectively, suggesting that significant improvement was achieved with the *a*TS-EVB model. Furthermore, as the orientational motion of water molecules within the hydrogen bond network plays a key role in PT,^{29,30} the rotational relaxation time (τ_2) for water molecules was calculated by integrating its reorientation correlation function

$$C_2(t) = \langle P_2[\mathbf{e}_{OH}(0) \cdot \mathbf{e}_{OH}(t)] \rangle, \quad (20)$$

where P_2 is the second Legendre polynomial and $\mathbf{e}$ is a unit vector along the OH bonds of a water molecule. The rotational relaxation time obtained with the *a*TS-EVB model is $\tau_2 = 2.65$ ps, which is in good agreement with the original *a*SPC/Fw water model³⁷ (2.7 ps) and the experimental data⁷³ (2.5 ps), providing an accurate description of the water orientational dynamics that leads to a significant improvement of the mobility of excess protons in water.

The PT mechanism was further analyzed by decomposing the total proton displacement into the continuous (vehicular) and discrete (Grothuss) contributions, using the same approach as that reported by Petersen and Voth.⁴⁸ As shown in Figure 6, the Grothuss mechanism accounts for $\sim 80\%$ of the total proton diffusion, which is in good agreement with the experimental effective Grothuss contribution of $\sim 79\%$, estimated from the Grothuss contribution D_G to the proton diffusion coefficient D_{H^+} at the infinite dilution. In this estimation, D_G is $\sim 0.74 \times 10^{-5} \text{ cm}^2/\text{s}$, as estimated via $D_G = D_{H^+} - D_{K^+}$, where $D_{K^+} = 0.198 \times 10^{-5} \text{ cm}^2/\text{s}$ is the experimental self-diffusion coefficient of the potassium cation that has an ionic radius similar to the hydronium ion, representing the vehicular contribution.³⁵

Using the *a*TS-EVB model, the classical activation energies E for the proton and water were calculated from the Arrhenius equation, $D = D_0 \exp(-E/RT)$, where D_0 is the constant value of $10^{-5} \text{ cm}^2/\text{s}$. Figure 7 shows the Arrhenius plots of the proton and water self-diffusion coefficients calculated from

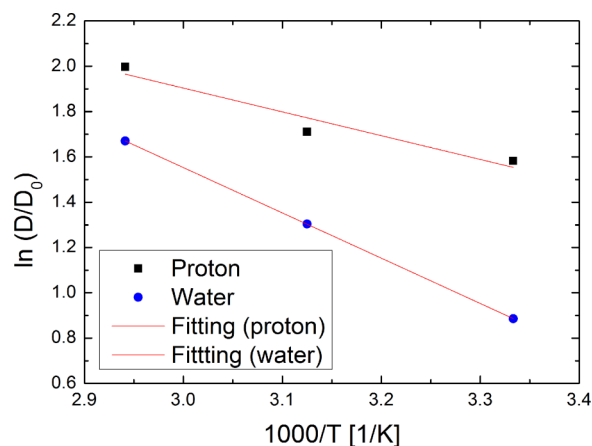


FIG. 7. Arrhenius plots of the proton and the water molecule. D_0 is the constant value of $10^{-5} \text{ cm}^2 \text{ s}^{-1}$. The red lines denote the linear fitting curves to the Arrhenius plots of the proton and water.

the corresponding MSD curves at temperatures of 298.15, 320, and 340 K. The E values calculated for the proton and water within this temperature range were found to be 0.090 eV and 0.17 eV, respectively, which are in excellent agreement with the experimental values of 0.093 eV and 0.17 eV.^{74,75} The activation energy of protons for the MS-EVB3 model,³⁵ in which the underlying water model is the nonpolarizable harmonic SPC/Fw model, also showed a good overall fit to the experimental values. As the diffusion constant is strongly affected by the strength of intermolecular hydrogen bonds,¹⁹ these agreements may be rationalized by the weaker hydrogen bond strength of the damped charge model for the α TS-EVB model and the charge delocalization effects for the MS-EVB3 model. This suggests that the damping is a good approximation of the electronic interaction between the solvated ions and water molecules, while simultaneously, dealing suitably with the electronic interaction is a key factor to obtain the correct value of the activation energy. Furthermore, electronic polarization effects can also contribute significantly to the dynamics of the water's hydrogen bond network.⁷⁶ The WK model, which takes account of electronic polarizability to reduce Franck–Condon barriers, resulted in the correct value of the activation energy, although the pre-exponential factor of proton mobility is approximately half of the experimental one. Kornyshev *et al.*³³ reported that using the WK model, the number of short jumps, where a proton jumps to only the nearest neighbors, is more than twice as great as the number of long jumps, where a proton jumps to the second nearest neighbors. Moreover, they suggest that their model overestimates the anticorrelated character of the successive directions of short jumps (i.e., the probability of short jumps in the opposite direction from the previous jump), which essentially results in a lower pre-exponential factor of the proton mobility. In this regard, the improvement of the pre-exponential factor in the α TS-EVB model may be attributed to the better agreement of the first valley and the second peak for $g_{\text{OH-O}}$ with experimental values, providing a better consecutive hydrogen bonding, which enhances the probability of short jumps in the same direction as the previous jump. For this reason, incorporating the polarization effects into the present model, in future, would be very important for improving the pre-exponential factor in the proton mobility with the conservation of the full charge of the ion, while retaining the same correct value of the activation energy and the proton solvation structure.

The proton hopping rate was calculated to investigate the correlation between the proton hopping events and the proton diffusion coefficients. Figure 8 shows the accumulation function of the average number of proton hops per proton at the temperatures 298.15, 320, and 340 K. As the functions were found to be linear overall, the proton hopping rates could then be calculated from the slopes of their linear fit. The rates were found to be 0.36, 0.42, and 0.49 ps⁻¹ for the respective temperatures. The results indicate that the hopping rate increases linearly as the temperature increases, suggesting a strong correlation with the proton diffusion coefficients. For the case at 298.15 K, assuming that the ratio of the experimental proton diffusion coefficient at 0.26M to infinite dilution is equal to that of the experimental effective hopping rate at 0.26M to infinite dilution (the experimental effective hopping rate at

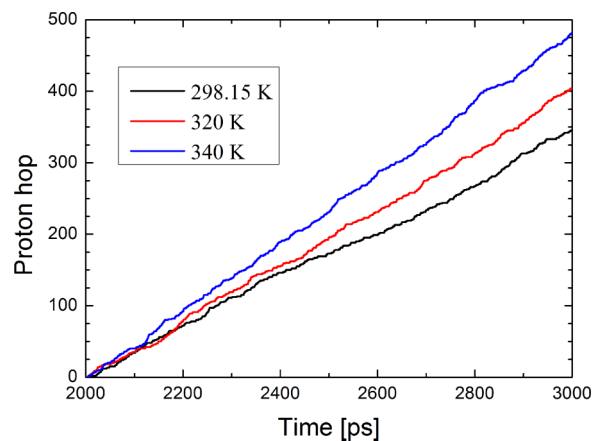


FIG. 8. Accumulation function of the average number of proton hops per proton at temperatures of 298.15, 320, and 340 K.

infinite dilution³⁵ is 0.71 ps⁻¹), the scaled value of 0.62 ps⁻¹ is obtained for the experimental effective hopping rate at 0.62M. In comparison with this hopping rate that obtained from the α TS-EVB model is roughly 40% lower than the experimental value, which is consistent with the difference between the proton self-diffusion coefficients for the α TS-EVB model and the experimental values, as mentioned earlier.

IV. CONCLUSIONS

We have examined the solvation structure and transport properties of protons in aqueous solutions to provide an accurate description of the excess proton mobility with the Grotthuss mechanism using classical MD. A modified version of the two-state EVB model, α TS-EVB, was developed based on the WK model to improve the quantitative description of both the proton and water mobility in aqueous environments. The new α TS-EVB model was a substantial improvement over the WK model in terms of the PT properties. This was achieved by modifying the following aspects: (i) the water force fields used in the α TS-EVB model are the nonpolarizable anharmonic water force fields α SPC/FW, as an alternative to the original polarizable water model, which provides pronounced nuclear quantum effects for anharmonic potentials and reproduces water properties much better and (ii) in the α TS-EVB model, the function $\Lambda(Q, R)$, an interaction matrix element in the Hamiltonian, was explicitly parameterized to reproduce the PESs calculated by the *ab initio* calculations for more reliable descriptions of the PT properties. The structural properties calculated from the RDFs are in good agreement with the experimental data obtained from the neutron and X-ray scattering measurements of the water structure factor, resulting in a significant improvement in water mobility over the WK model. The RDF between the excess protons and the water molecules also yields the main feature of the experimental data, contributing to the improvement of the interaction between the hydronium ions and water molecules. The α TS-EVB model improves a description of PT via the Grotthuss hopping mechanism, giving a significantly enhanced transport, in comparison with the WK model. Specifically, given the proton concentration in the HCl solution calculated in this study, the

self-diffusion coefficient is found to be within $\sim 33\%$ of the experimental value, giving a significantly enhanced transport. The classical activation energies for the proton and water were calculated from the Arrhenius equation between 298.15 and 340 K (the temperature range used in this study). The values calculated from the linear fit to the Arrhenius plots are found to be 0.090 eV and 0.17 eV for the proton and water, respectively, which are in excellent agreement with the experimental values of 0.093 eV and 0.17 eV. Although the *a*TS-EVB model underestimates the value of the hopping rate compared with that of the experimental values, the ratio between them is consistent with the difference between the diffusion coefficients for the *a*TS-EVB model and the experimental values, suggesting their strong correlation.

Besides now being able to carry out simpler and faster calculations, with the new *a*TS-EVB model, it is now possible to obtain an accurate estimation of PT properties including the prediction of the absolute values of mobility within the simplicity of the theoretical framework, hence establishing an important link between proton mobility and its surrounding structures and environments. In this regard, the *a*TS-EVB model could attract wider usage and facilitate the efficient application of the PT phenomena to other longer simulations and larger systems in highly concentrated and confined heterogeneous multiproton environments, such as in polymer electrolyte membranes and along water wires in narrow pores.

ACKNOWLEDGMENTS

This work was supported by the New Energy and Industrial Technology Development Organization (NEDO) of Japan and the Japan Society for the Promotion of Science (JSPS) Grant-in-Aid for JSPS Fellows, Grant No. 25-9722. It was performed, in part, using the supercomputer of the Institute of Fluid Science (IFS) of Tohoku University.

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